

Sri Lanka Institute of Information Technology



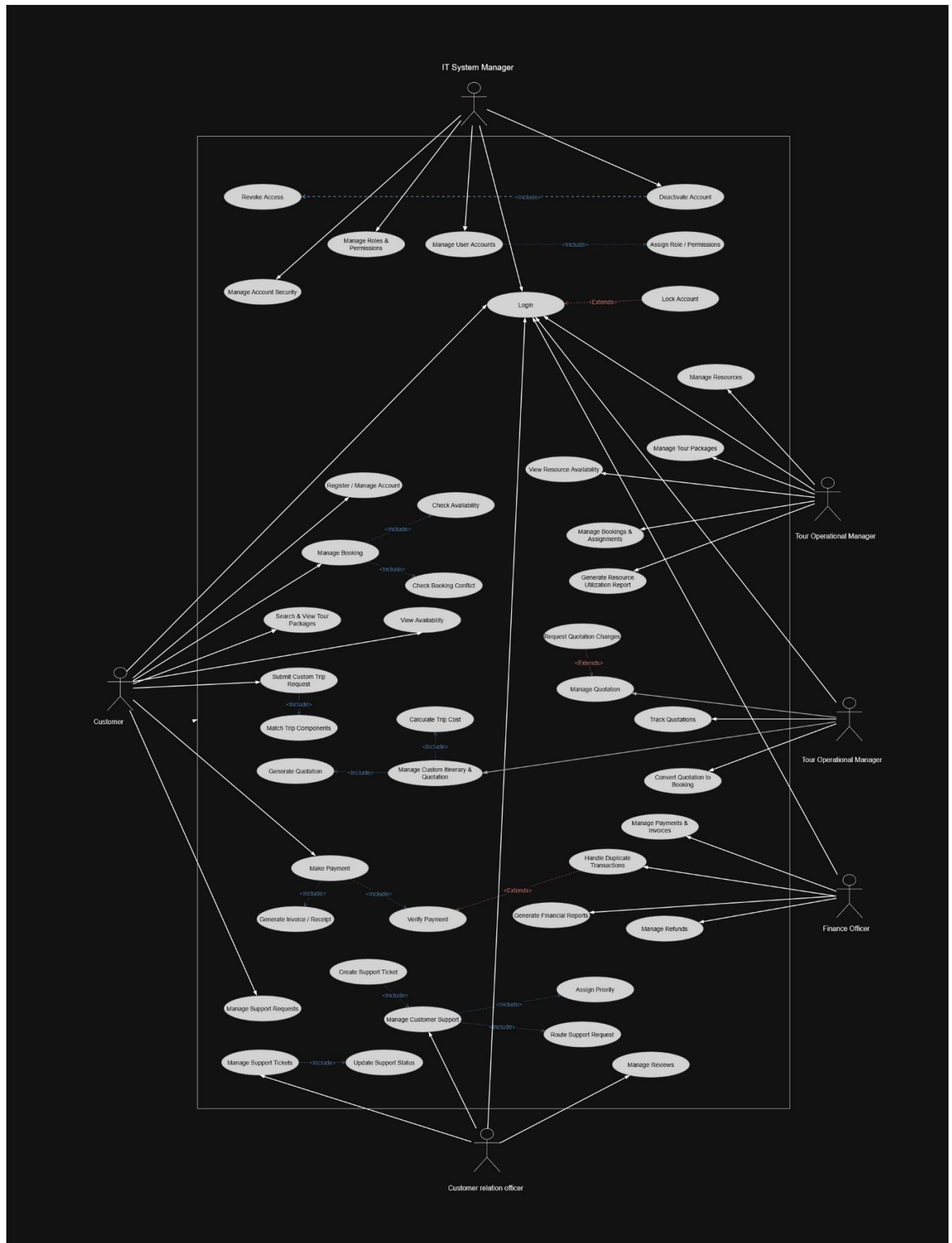
Lab Submission - 03 <2026-Y2-S1-KU-08 >

Member	Student ID	Assigned Requirement
K.J. Jaswinth	IT25100569	Tour & Package Management
M.H.M. Basith	IT25100552	Booking & Resource Availability Management
M.K.P.P Ranawake	IT25100561	Trip planning & Quotation Management
H.M.M.P Herath	IT25100559	Payment Processing & Invoicing
S.M.L.M.B Senanayake	IT25100585	Customer Communication & Support
H.M.T.D Herath	IT25100644	User & Account Management

<Island Trails Lanka (Pvt) Ltd> <Web-Based Tour and Tourism Management System>

Software Engineering | SE2030
B.Sc. (Hons) in Information Technology

1. User Case Diagram



2. User Case Scenarios

2.1 TOUR & PACKAGE MANAGEMENT (K.J. Jaswinth - IT25100569)

Number	1	
Name	Add and manage tour packages	
Summary	Tour Operations Manager or Admin adds a new tour package to the system or updates an existing package's details, ensuring the catalogue is always current and accurate.	
Priority	5	
Preconditions	Tour Operations Manager is logged in with staff privileges; they have permission to create or edit packages	
Postconditions	A new or updated tour package exists in the system and is immediately visible to customers in search results	
Primary Actor(s)	Tour Operations Manager	
Trigger	Tour Operations Manager selects "Add New Package" or "Edit Package" in the admin portal	
Main Scenario	#	Action
	1	System displays the package creation/edit form with fields for name, destination, duration, itinerary details, pricing, accommodation options, transport type, and images
	2	Tour Operational Manager enters or updates the package information
	3	System validates that all required fields are filled and pricing is numeric and positive
	4	System creates or updates the package record
Extensions	#	Branchin g Action
	3a	Required fields are missing or invalid — system displays validation errors and returns focus to the form
Open Issue	#	Issue
	1	How long should a "Draft" package be retained in the system before auto-deletion?

2.2 BOOKING & RESOURCE AVAILABILITY MANAGEMENT (M.H.M. Basith - IT25100552)

Number		2
Name		Create and manage Booking for Tour, Hotel, or Transport
Summary		Customer books a tour, hotel, or transport service online by selecting a package and confirming resource assignments. The system checks real-time availability to prevent double-booking and assigns the booking a status so the customer can track its progress.
Priority		5
Preconditions		Customer is logged into the system; a tour, hotel, or transport package exists and is available for the requested dates
Postconditions		A new booking record is created with status "Pending Payment", resource assignments are confirmed, and the customer receives a booking confirmation email with booking ID
Primary Actor		Customer
Secondary Actor		Tour Operations Manager
Trigger		Customer selects a package and clicks "Book Now"
Main Scenario	#	Action
	1	System displays the selected package details and a form requesting customer information
	2	Customer enters their information and selects the desired travel dates
	3	System checks real-time availability of all resources
	4	System finds all resources available — no conflicts detected
	5	Customer confirms the booking, and system advances to the payment step
	6	System sends the customer a confirmation email with booking ID, package details, and payment instructions
	7	Tour Operations Manager receives an alert that a new booking has been created and added to the dashboard
Extensions	#	Branching Action
	3a	Conflict detected — one or more resources are already assigned to another booking for the selected dates
	3a.1	System displays the conflicting resource(s) and suggests alternative dates or alternative resources of the same type
	3a.2	Customer selects an alternative date or resource, and process resumes at step 3
	3a.3	No alternative is acceptable — customer cancels the booking; system returns to package search
Open Issue	#	Issue
	1	Should a partial-overlap (e.g. 15-minute buffer between a drop-off and the next pickup) count as a conflict, or only a true time-range overlap?

2.3 TRIP PLANNING & QUOTATION MANAGEMENT (M.K.P.P Ranawake - IT25100561)

Number		3
Name		Submit Custom Trip Request and Receive Quotation
Summary		Customer submits a custom trip request with budget, travel dates, group size, and interests. Travel Consultant matches available destinations, hotels, and guides, generates an itinerary and system calculates the total cost, and sends a quotation to the customer for review
Priority		4
Preconditions		A quotation exists with status "Accepted" by the customer
Postconditions		confirmed booking exists containing the itinerary, resources, and price from the accepted quotation
Primary Actor		Travel Consultant
Trigger		Travel Consultant selects "Convert to Booking" on an accepted quotation
Main Scenario	#	Action
	1	System displays the accepted quotation with its itinerary, resources, and total cost
	2	Travel Consultant confirms the conversion
	3	System checks the requested resources for scheduling conflicts
	4	System copies the quotation's itinerary, resource assignments, and price into a new booking record
	5	System sets the quotation status to "Converted" and the new booking status to "Confirmed"
	6	System sends a booking confirmation alert to the customer
Extensions	#	Branching Action
	3a	A conflict is detected on one or more resources - system halts the conversion and displays the conflicting resource to the Travel Consultant
	3a.1	Travel Consultant selects an alternative available resource for the conflicting item
	3a.2	Process resumes at step 5 with the updated resource assignment
Open Issue	#	Issue
	1	If a quotation is not converted within a set number of days after acceptance, should it expire and require re-quoting?

2.4 PAYMENT PROCESSING & INVOICING (H.M.M.P Herath - IT25100559)

Number		4
Name		Process Payment and Generate Invoice
Summary		Customer completes payment for a booking using a payment method, generates an invoice and digital receipt, updates the booking status to "Confirmed"
Priority		5
Preconditions		the customer has proceeded to the payment page
Postconditions		Payment is marked as "Verified", booking status is updated to "Confirmed", invoice and receipt are generated and sent to customer, and Finance Officer receives notification
Primary Actor		Finance & Administration Officer
Secondary Actor		Customer
Trigger		The payment gateway returns a transaction result for a submitted payment
Main Scenario	#	Action
	1	System displays the payment form
	2	Customer selects a payment method and enters their payment details.
	3	System validates the payment information
	4	System checks the transaction ID against existing verified transactions to prevent duplicate charging
	5	System generates an invoice/receipt for the customer.
	6	System notifies the Finance & Administration Officer and the customer that payment was successfully verified
Extensions	#	Branching Action
	2a	Payment information is invalid
	4a	System detects a duplicate transaction ID
	4a.1	System notifies the Finance & Administration Officer to investigate before the customer is charged again
Open Issue	#	Issue
	1	Should failed/timed-out payment attempts be retained in the transaction log, or discarded after a set retry window?

2.5 CUSTOMER COMMUNICATION & SUPPORT (S.M.L.M.B Senanayake - IT25100585)

Number	5	
Name	Manage inquiries and customer reviews	
Summary	Customer submits an inquiry ,complaint or review through the system. The system creates a trackable ticket, assigns it to the appropriate department, and sends the customer an acknowledgement with a ticket ID for status tracking	
Priority	4	
Preconditions	A customer inquiry or complaint has been received through a supported channel	
Postconditions	A ticket exists with a unique ID, an initial status of "Open", and is routed to the correct department	
Primary Actor	Customer Relations Officer	
Secondary Actor	Customer	
Trigger	A new inquiry or complaint is received	
Main Scenario	#	Action
	1	System captures the inquiry
	2	System creates a new ticket record with status "Open" and a unique ticket ID
	3	System assigns the ticket to the correct department and priority level automatically
	4	System displays the newly created, routed ticket in the Customer Relations Officer's inbox
	5	System sends the customer an acknowledgement with the ticket ID
Extensions	#	Branchin g Action
	3a	The routing rules cannot confidently match the ticket to a department system routes it to a default "General Inquiries" queue and flags it for manual reassignment
Open Issue	#	Issue
	1	Should a customer be able to reply to their own ticket and add further messages, or must all follow-up go through a new ticket?

2.6 USER & ACCOUNT MANAGEMENT (H.M.T.D Herath - IT25100644)

Number	6	
Name	Register and Manage User Account	
Summary	Allows a new user to register an account and log in to the system, and allows the IT System Manger to manage user accounts, roles, and permissions.	
Priority	5	
Preconditions	IT Systems Officer is logged in with admin privileges	
Postconditions	A new account is created and stored in the system, or an existing account's profile, role, or status is successfully updated.	
Primary Actor	IT Systems Officer	
Trigger	IT Systems Officer selects "Create Staff Account" in the admin panel	
Main Scenario	#	Action
	1	System displays the staff account creation form with a role dropdown
	2	IT Systems Officer enters the employee's name, email, and selects a role
	3	System validates that all required fields are filled and the email is not already registered
	4	System creates the staff account, assigns the selected role's permission set, and sets the account status to "Active"
	5	System displays confirmation to the IT Systems Officer
Extensions	#	Branchin g Action
	3a	The email is already registered to another account - system displays an error and prevents account creation until a different email is entered
	3b	A required field is missing or invalid - system displays a validation error and returns the IT Systems Officer to the form
Open Issue	#	Issue
	1	Should newly created accounts require the employee to set their own password on first login, rather than receiving one by email?